

Karthik Vishal S

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EDUCATION

Amrita Vishwa Vidyapeetham

BTech in Computer Science Engineering; CGPA: 8.53

Thamarai International School

Senior Secondary – Boards: 89% JEE Mains: 96.99%ile

Coimbatore, Tamil Nadu

Aug. 2023 – July 2027

Thanjavur, Tamil Nadu

Aug. 2021 – May 2023

EXPERIENCE

Fidelity Investments - LEAP intern

Chennai, Tamil Nadu

May 2026 – July 2026

- Selected for the prestigious LEAP (Learning Early Assistance Program), collaborating with cross-functional teams on enterprise-level financial technology solutions.

Full-Stack Development Freelancer

June 2025 – Present

- Developed a production-grade **E-commerce platform** for a boutique integrating Stripe for secure payments, implementing authentication, bcrypt hashing, and Nodemailer for password recovery and developed a **Patient Management System** for an Orthopaedician to manage patient details and prescription generations .

Head Coordinator (Tech-Fair) - Anokha '26

October 2025 – January 2026

- Curated and shortlisted a premier selection of Computer Science projects for a high-profile Departmental Expo, presenting innovative solutions to an elite panel of 15+ CEOs and over 75 industry delegates.

MAJOR PROJECTS

Soundaryas Boutique | *NextJs, NextAuth, MongoDB, Stripe, TailwindCSS, MaterialUI, ExpressJs, Vercel*

- Built a scalable MERN-stack e-commerce application with modular architecture, secure authentication, and RESTful APIs for product and order management.
- Optimized frontend using React.js + Tailwind CSS for responsive UI, and implemented efficient state management to handle real-time cart and checkout operations.

LithoFlow - Smart Water Bottle | *ESP32, TinyML, Next.js, MongoDB, C++, TensorFlow Lite, Node.js*

- Developed an IoT Level-5 device using a 24-bit ADC to track fluid intake with **98%** precision and Deployed a quantized Neural Network (TinyML) on ESP32, reducing edge footprint to **2 KB**.
- Built a Next.js API with **ngrok** to sync telemetry to MongoDB with sub-**500ms** latency. Implemented EEPROM persistence and a Python engine to predict hydration gaps for doctors.

Flash Sale Engine | *Redis, LuaScript, AWS SQS, NodeJs, NextJs, ApacheJmeter*

- A highly resilient, distributed architecture that uses Redis atomic locking and **AWS SQS buffering** to achieve zero oversells under extreme load, guaranteeing data integrity during flash sales.

Fabergé | *NextJs, FlaskAPI, Gemini Flash, SupaBase, TailwindCSS, MaterialUI, Railway, ReCharts, Vercel*

- Fabergé uses advanced AI to create personalized Spotify playlists based on your specific feelings and style. Our platform combines deep data insights with music industry standards to deliver a premium listening experience.

Clause AI | *NextJS, FastAPI, SupaBase, TailwindCSS, Gemini API, NLTK*

- A Generative AI app that is designed to simplify and analyze legal documents by leveraging AI.
- Implemented an interactive Chatbot that answers all questions related to the simplified clauses

TECHNICAL SKILLS & CERTIFICATIONS

Languages: JavaScript, Python, C++, C, SQL, HTML/CSS

Frontend: React, Next.js, TailwindCSS, MaterialUI, Firebase, Bootstrap, ShadCN

Backend: Node.js, Express.js, MongoDB, PostgreSQL, SupaBase, Firebase, Cloudinary

Tools: GitHub, Vercel, VS Code, Eclipse, Streamlit, Postman, Apache JMeter

Certifications:

META Backend Developer - Coursera

The Complete FullStack Web Development - Udemy